MA1522 Linear Algebra for Computing
L2 Lesson 15



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QR Factorization

Let $\boldsymbol{A}$ be an $m \times n$ matrix, and suppose that $\operatorname{rank}(\boldsymbol{A}) = n$, or equivalently, the columns of $\boldsymbol{A}$ are linearly independent.

A **QR factorization of A** is a decomposition of $\boldsymbol{A}$ into a product

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R},$$

where

- $\boldsymbol{Q}$ is an $m \times n$ matrix with **orthonormal columns**;
- $\boldsymbol{R}$ is an $n \times n$ **upper triangular matrix** with **positive diagonal entries**.

Note

Recall that $\boldsymbol{Q}$ has orthonormal columns if and only if $\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}_n$.

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Recall that Q has orthonormal columns if and only if $Q^T Q = I_n$.

QR Factorization

Let $\boldsymbol{A}$ be an $m \times n$ matrix, and suppose that $\operatorname{rank}(\boldsymbol{A}) = n$, or equivalently, the columns of $\boldsymbol{A}$ are linearly independent.

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Note

Recall that $\boldsymbol{Q}$ has orthonormal columns if and only if $\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}_n$.

Computing QR Factorization

Let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{c}_1 & \boldsymbol{c}_2 & \cdots & \boldsymbol{c}_n \end{pmatrix}$ with $\operatorname{rank}(\boldsymbol{A}) = n$. Then $\boldsymbol{c}_1, \boldsymbol{c}_2, \ldots, \boldsymbol{c}_n$ are linearly independent. Apply Gram-Schmidt process to $\boldsymbol{c}_1, \boldsymbol{c}_2, \ldots, \boldsymbol{c}_n$:

$$\begin{aligned}
\boldsymbol{v}_1 &= \boldsymbol{c}_1, \\
\boldsymbol{v}_2 &= \boldsymbol{c}_2 - \frac{\boldsymbol{c}_2 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2} \boldsymbol{v}_1, \\
\boldsymbol{v}_3 &= \boldsymbol{c}_3 - \frac{\boldsymbol{c}_3 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2} \boldsymbol{v}_1 - \frac{\boldsymbol{c}_3 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2} \boldsymbol{v}_2, \\
&\vdots \\
\boldsymbol{v}_k &= \boldsymbol{c}_k - \frac{\boldsymbol{c}_k \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2} \boldsymbol{v}_1 - \frac{\boldsymbol{c}_k \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2} \boldsymbol{v}_2 - \cdots - \frac{\boldsymbol{c}_k \cdot \boldsymbol{v}_{k-1}}{\|\boldsymbol{v}_{k-1}\|^2} \boldsymbol{v}_{k-1}.
\end{aligned}$$

Let $\boldsymbol{Q} = \begin{pmatrix} \frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1 & \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2 & \cdots & \frac{1}{\|\boldsymbol{v}_n\|}\boldsymbol{v}_n \end{pmatrix}$ and $\boldsymbol{R} = (r_{ij})_{n \times n}$ where

$$r_{ij} = \begin{cases} \|\boldsymbol{v}_i\|, & \text{if } i = j; \\ \frac{\boldsymbol{c}_j \cdot \boldsymbol{v}_i}{\|\boldsymbol{v}_i\|} = \frac{\boldsymbol{c}_j \cdot \boldsymbol{v}_i}{\|\boldsymbol{v}_i\|^2} \|\boldsymbol{v}_i\|, & \text{if } i < j; \\ 0, & \text{if } i > j. \end{cases}$$

Then $\boldsymbol{Q}$ has orthonormal columns, $\boldsymbol{R}$ is upper triangular with positive diagonal entries, and $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$.

Computing QR Factorization

Let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{c}_1 & \boldsymbol{c}_2 & \cdots & \boldsymbol{c}_n \end{pmatrix}$ with $\operatorname{rank}(\boldsymbol{A}) = n$. Then $\boldsymbol{c}_1, \boldsymbol{c}_2, \ldots, \boldsymbol{c}_n$ are linearly independent. Apply Gram-Schmidt process to $\boldsymbol{c}_1, \boldsymbol{c}_2, \ldots, \boldsymbol{c}_n$:

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\boldsymbol{v}_1 &= \boldsymbol{c}_1, \\
\boldsymbol{v}_2 &= \boldsymbol{c}_2 - \tfrac{\boldsymbol{c}_2 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2} \boldsymbol{v}_1, \\
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&\vdots \\
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\end{aligned}$$

Let $\boldsymbol{Q} = \begin{pmatrix} \frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1 & \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2 & \cdots & \frac{1}{\|\boldsymbol{v}_n\|}\boldsymbol{v}_n \end{pmatrix}$ and $\boldsymbol{R} = (r_{ij})_{n \times n}$ where

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Then $\boldsymbol{Q}$ has orthonormal columns, $\boldsymbol{R}$ is upper triangular with positive diagonal entries, and $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$.

Computing QR Factorization

Remarks

When $\boldsymbol{A} = \boldsymbol{QR}$ is a QR factorization of $\boldsymbol{A}$, we have

$$\boldsymbol{Q}^T\boldsymbol{A} = \boldsymbol{Q}^T(\boldsymbol{QR}) = (\boldsymbol{Q}^T\boldsymbol{Q})\boldsymbol{R} = (\boldsymbol{I}_n)\boldsymbol{R} = \boldsymbol{R}.$$

Thus one can also compute $\boldsymbol{R}$ from the knowledge of $\boldsymbol{A}$ and $\boldsymbol{Q}$ in this way.

Question 1

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Find QR factorizations of the following matrices: [Pg 54-55 of Chapter 5]

a. $\begin{pmatrix} 0 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

b. $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

Question 1a: Solution

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\begin{aligned}
\boldsymbol{v}_1 &= (0,1,1,1)^T && = (0,1,1,1)^T, \\
\boldsymbol{v}_2 &= (1,-1,0,1)^T - \tfrac{0}{3}(0,1,1,1)^T && = (1,-1,0,1)^T, \\
\boldsymbol{v}_3 &= (1,1,1,1)^T - \tfrac{3}{3}(0,1,1,1)^T - \tfrac{1}{3}(1,-1,0,1)^T && = \tfrac{1}{3}(2,1,0,-1)^T.
\end{aligned}$$

Thus,

$$\boldsymbol{Q} = \begin{pmatrix} 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{pmatrix}, \qquad \boldsymbol{R} = \begin{pmatrix} \sqrt{3} & 0 & \sqrt{3} \\ 0 & \sqrt{3} & \frac{1}{\sqrt{3}} \\ 0 & 0 & \sqrt{\frac{2}{3}} \end{pmatrix}.$$

Question 1b: Solution

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\begin{aligned}
\boldsymbol{v}_1 &= (1,1,1,0)^T && = (1,1,1,0)^T, \\
\boldsymbol{v}_2 &= (1,1,1,1)^T - \tfrac{3}{3}\,(1,1,1,0)^T && = (0,0,0,1)^T, \\
\boldsymbol{v}_3 &= (0,0,1,1)^T - \tfrac{1}{3}(1,1,1,0)^T - \tfrac{1}{1}(0,0,0,1)^T && = \tfrac{1}{3}(-1,-1,2,0)^T.
\end{aligned}$$

Thus,

$$\boldsymbol{Q} = \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ 0 & 1 & 0 \end{pmatrix}, \qquad \boldsymbol{R} = \begin{pmatrix} \sqrt{3} & \sqrt{3} & \frac{1}{\sqrt{3}} \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{\frac{2}{3}} \end{pmatrix}.$$

Least Squares Solution

Let $\boldsymbol{A}$ be an $m \times n$ matrix and let $\boldsymbol{b} \in \mathbb{R}^m$.

A **least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$** is a vector $\boldsymbol{u} \in \mathbb{R}^n$ such that

$$\|\boldsymbol{Au} - \boldsymbol{b}\| \leq \|\boldsymbol{Av} - \boldsymbol{b}\|$$

for all $\boldsymbol{v} \in \mathbb{R}^n$.

Recall that

$$\mathrm{Col}(\boldsymbol{A}) = \text{the span of columns of } \boldsymbol{A} = \{\boldsymbol{Ax} \mid \boldsymbol{x} \in \mathbb{R}^n\}.$$

Theorem

The following statements are equivalent:

1. $\boldsymbol{u} \in \mathbb{R}^n$ *is a least squares solution of* $\boldsymbol{Ax} = \boldsymbol{b}$.
2. $\boldsymbol{Au}$ *is the orthogonal projection of* $\boldsymbol{b}$ *onto* $\mathrm{Col}(\boldsymbol{A})$.
3. $\boldsymbol{A}^T\boldsymbol{Au} = \boldsymbol{A}^T\boldsymbol{b}$.

Remarks

$\boldsymbol{Ax} = \boldsymbol{b}$ is consistent

$\Leftrightarrow \boldsymbol{b} \in \mathrm{Col}(\boldsymbol{A})$

$\Leftrightarrow$ orthogonal projection of $\boldsymbol{b}$ onto $\mathrm{Col}(\boldsymbol{A})$ equals $\boldsymbol{b}$

$\Leftrightarrow \boldsymbol{Au} = \boldsymbol{b}$ for all least squares solutions $\boldsymbol{u}$ of $\boldsymbol{Ax} = \boldsymbol{b}$.

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The following statements are equivalent:

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$\Leftrightarrow \boldsymbol{Au} = \boldsymbol{b}$ for all least squares solutions $\boldsymbol{u}$ of $\boldsymbol{Ax} = \boldsymbol{b}$.

Question 2

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Let $\boldsymbol{A} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} 6 \\ 3 \\ -1 \\ 1 \end{pmatrix}$.

a. Find all least squares solutions of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.

b. Use the result in (a) to find the orthogonal projection of $\boldsymbol{b}$ onto the column space of $\boldsymbol{A}$.

c. Is $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ consistent?

[Pg 65–68 of Chapter 5]

Question 2

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c. Is $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ consistent?

[Pg 65–68 of Chapter 5]

Solution.

a. $(\boldsymbol{A}^T\boldsymbol{A} \mid \boldsymbol{A}^T\boldsymbol{b}) = \left(\begin{array}{cccc|c} 3 & 0 & 3 & 0 & 3 \\ 0 & 3 & 1 & 2 & 4 \\ 3 & 1 & 4 & 0 & 9 \\ 0 & 2 & 0 & 2 & -2 \end{array}\right) \xrightarrow[\text{Jordan}]{\text{Gauss-}} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & -6 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & 7 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$.

Let $x_4 = t$. Then $x_1 = -6 - t$, $x_2 = -1 - t$, $x_3 = 7 + t$. So

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -6-t \\ -1-t \\ 7+t \\ t \end{pmatrix}, \quad t \in \mathbb{R}.$$

Question 2(b,c): Solution

b. From (a), $\boldsymbol{u} = \begin{pmatrix} -6 \\ -1 \\ 7 \\ 0 \end{pmatrix}$ is a least squares solution of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. Thus, the orthogonal projection of $\boldsymbol{b}$ onto $\mathrm{Col}(\boldsymbol{A})$ equals

$$\boldsymbol{A}\boldsymbol{u} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -6 \\ -1 \\ 7 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 1 \\ 0 \end{pmatrix}.$$

c. From (b), the orthogonal projection of $\boldsymbol{b}$ is not equal to $\boldsymbol{b}$, so the equation $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ is not consistent.

Question 2(b,c): Solution

b. From (a), $\boldsymbol{u} = \begin{pmatrix} -6 \\ -1 \\ 7 \\ 0 \end{pmatrix}$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$. Thus, the orthogonal projection of $\boldsymbol{b}$ onto $\mathrm{Col}(\boldsymbol{A})$ equals

$$\boldsymbol{Au} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -6 \\ -1 \\ 7 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 1 \\ 0 \end{pmatrix}.$$

c. From (b), the orthogonal projection of $\boldsymbol{b}$ is not equal to $\boldsymbol{b}$, so the equation $\boldsymbol{Ax} = \boldsymbol{b}$ is not consistent.

Question 3

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Let $\boldsymbol{A}$ be a $m \times n$ matrix with $\text{rank}(\boldsymbol{A}) = n$, and let $\boldsymbol{A} = \boldsymbol{QR}$ be the QR factorization of $\boldsymbol{A}$. Let $\boldsymbol{b} \in \mathbb{R}^m$. Which of the following statements is/are true?

a. If $\boldsymbol{u} \in \mathbb{R}^n$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$, then $\boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b}$.

b. If $\boldsymbol{u} \in \mathbb{R}^n$ satisfies $\boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b}$, then $\boldsymbol{u}$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$.

c. $\boldsymbol{Ax} = \boldsymbol{b}$ has a unique least squares solution.

Question 3

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a. If $\boldsymbol{u} \in \mathbb{R}^n$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$, then $\boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b}$.

. . . . . **Ans: True**

b. If $\boldsymbol{u} \in \mathbb{R}^n$ satisfies $\boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b}$, then $\boldsymbol{u}$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$.

c. $\boldsymbol{Ax} = \boldsymbol{b}$ has a unique least squares solution.

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b. If $\boldsymbol{u} \in \mathbb{R}^n$ satisfies $\boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b}$, then $\boldsymbol{u}$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$.

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c. $\boldsymbol{Ax} = \boldsymbol{b}$ has a unique least squares solution.

Question 3

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Let $\boldsymbol{A}$ be a $m \times n$ matrix with $\text{rank}(\boldsymbol{A}) = n$, and let $\boldsymbol{A} = \boldsymbol{QR}$ be the QR factorization of $\boldsymbol{A}$. Let $\boldsymbol{b} \in \mathbb{R}^m$. Which of the following statements is/are true?

a. If $\boldsymbol{u} \in \mathbb{R}^n$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$, then $\boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b}$.
. . . . . **Ans: True**

b. If $\boldsymbol{u} \in \mathbb{R}^n$ satisfies $\boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b}$, then $\boldsymbol{u}$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$.
. . . . . **Ans: True**

c. $\boldsymbol{Ax} = \boldsymbol{b}$ has a unique least squares solution. . . . . **Ans: True**

Question 3: Solution

$\boldsymbol{u} \in \mathbb{R}^n$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$

$\Leftrightarrow \boldsymbol{A}^T\boldsymbol{Au} = \boldsymbol{A}^T\boldsymbol{b}$

$\Leftrightarrow (\boldsymbol{R}^T\boldsymbol{Q}^T)(\boldsymbol{QR})\boldsymbol{u} = \boldsymbol{R}^T\boldsymbol{Q}^T\boldsymbol{b}$

$\Leftrightarrow \boldsymbol{R}^T\boldsymbol{Ru} = \boldsymbol{R}^T\boldsymbol{Q}^T\boldsymbol{b}$

$\Leftrightarrow \boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b}$ since $\boldsymbol{R}^T$ is invertible (as $\det(\boldsymbol{R}^T) = \det(\boldsymbol{R}) > 0$)

$\Leftrightarrow \boldsymbol{u} = \boldsymbol{R}^{-1}\boldsymbol{Q}^T\boldsymbol{b}$.

Question 4

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Recall from Question 1(b) that

$$\boldsymbol{A} := \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{3} & \sqrt{3} & \frac{1}{\sqrt{3}} \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{\frac{2}{3}} \end{pmatrix}$$ is a QR factorization of $\boldsymbol{A}$.

Find the least squares solution of $\boldsymbol{A}\boldsymbol{x} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$. [Use Qn 3]

Question 4

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Recall from Question 1(b) that

$$\boldsymbol{A} := \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{3} & \sqrt{3} & \frac{1}{\sqrt{3}} \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{\frac{2}{3}} \end{pmatrix}$$ is a QR factorization of $\boldsymbol{A}$.

Find the least squares solution of $\boldsymbol{Ax} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$. [Use Qn 3]

Solution.

$$\begin{pmatrix} \sqrt{3} & \sqrt{3} & \frac{1}{\sqrt{3}} \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{\frac{2}{3}} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \boldsymbol{Ru} = \boldsymbol{Q}^T\boldsymbol{b} = \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ 0 & 1 & 0 \end{pmatrix}^T \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ 0 \\ -\frac{1}{\sqrt{6}} \end{pmatrix}.$$

Thus, $\boldsymbol{u} = (0, \frac{1}{2}, -\frac{1}{2})^T$.

□

Next Lesson ...

We will explore **Section 6.1**.
Please watch/review the video clip for this section before then.